

CYBER SECURITY

How It Affects Businesses



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SYSTEMATIC LITERATURE REVIEW

- Our search was limited to texts that were published from 2010 to 2019.
- We gathered 32 papers
- Using accuracy and the prestige of the papers with the help of search machines - like google scholar- we selected information that enhanced our knowledge and gave our presentation scientific accuracy.



Using journal and conference papers alike, we were able to

SYSTEMATIC LITERATURE REVIEW

The literature review was conducted following the Kitchenham (2004) model, which consists of three phases:

- Development of a review protocol that includes the objectives, methodology, and expected outcomes.
- Conducting the review by searching databases, applying inclusion/exclusion criteria, and reviewing the results.
- Communicating the results through a structured report of findings.



PHASES OF THE LITERATURE REVIEW

PHASE 1

Planning the Review

- **Formulation of research questions**
- **Selection of databases**
- **Definition of inclusion/exclusion criteria**
- **Data extraction strategy**
- **Quality assessment plan**

PHASE 2

Conducting the Review

- **Identification of relevant studies**
- **Selection based on criteria**
- **Data extraction from papers**
- **Synthesis of findings**
- **Quality assessment of studies**

PHASE 3

Reporting the Review

- **Communication of results**
- **Discussion of principal findings**
- **Identification of limitations**
- **Recommendations for practice**
- **Future research directions**



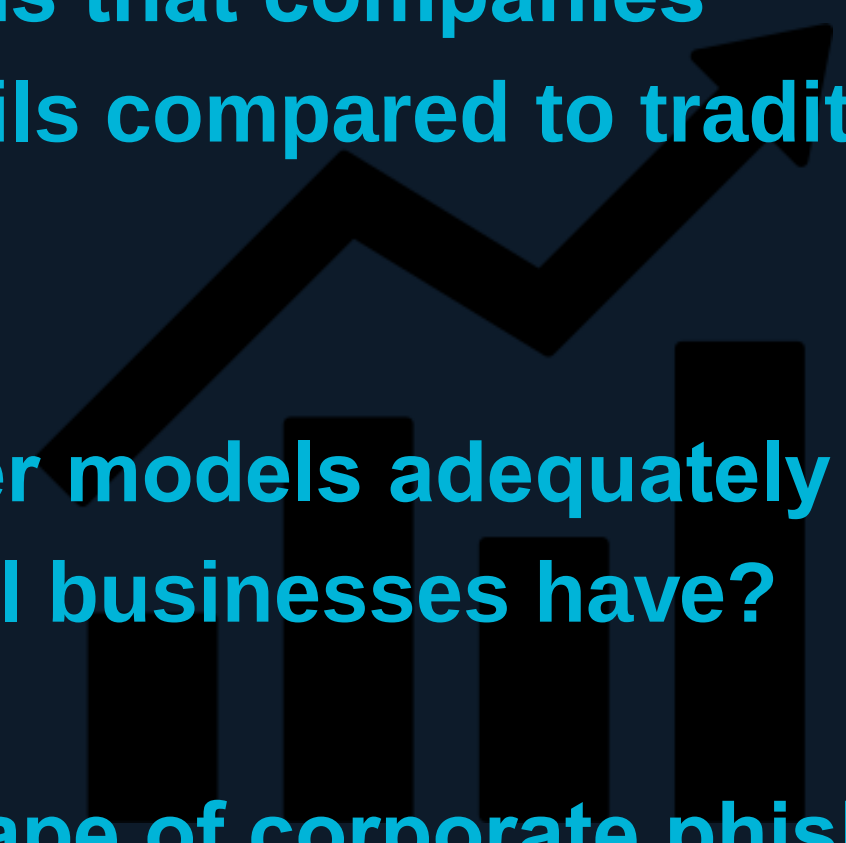
Research questions - Analysis

In this paper, we analyzed the impact that cybersecurity has on businesses on multiple levels.

In this presentation we will discuss about the financial impact, the humanity aspect, the vulnerability of small enterprises, the risk of emerging technologies and lastly the legal conditions and the various mechanisms used to protect themselves.



OUR RESEARCH QUESTIONS

- **RQ1: How do major data breaches affect stock values and brand equality of publicly traded companies?**
 - **RQ2: How effective are those cyber security training programs that companies incorporate in reducing employee click-rates on phishing emails compared to traditional training?**
 - **RQ3: Are the current managed cyber security service provider models adequately serving their purpose and meeting the special needs that small businesses have?**
 - **RQ4: How does the adoption of AI affect and alter the landscape of corporate phishing and social engineering tasks?**
 - **RQ5: How have strict data privacy laws (like GDPR or CCPA) shifted and changed corporate cyber security spending and global data strategies?**
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DATA BREACHES & STOCK VALUES

Stocks drop significantly within days of a breach disclosure

Volatility stays elevated — investors demand higher risk compensation

Financial and tech firms suffer the steepest market reactions

Brand equity erodes fast — Equifax lost ~\$4B in one week

Recovery takes 45–60 trading days depending on sector

Cybersecurity spending is a risk-reduction investment, not a sunk cost

REDUCING PHISHING VULNERABILITY

Phishing exploits human weaknesses — stress, fatigue, and risk-taking increase clicks

Continuous simulations beat annual training — lower click rates and better retention

Deep-learning models (Transformers) detect phishing at ~97% accuracy

DevSecOps approach embeds security early and improves user engagement

Strong security culture at work transfers good habits to home behavior

Proposed HCPD framework unites AI detection, psychology profiling, and governance

SME CYBER VULNERABILITY

SMEs are 90%+ of businesses but lack budget, staff, and awareness for cybersecurity

Phishing and ransomware exploit untrained staff — insider threats worsen without policies

Security ROI is invisible until a breach — SMEs stay reactive, not proactive

58% of attacks hit SMEs to infiltrate larger supply chains — a systemic risk

MSSPs are built for enterprises — too complex and expensive for small businesses

Solution: government-supported, SME-tailored services prioritizing core protections

AI & CORPORATE PHISHING

AI generates personalized phishing at scale — 82.6% of phishing emails in 2025 showed AI content

LLMs produce perfect grammar, making old red flags obsolete — deepfakes clone executive voices

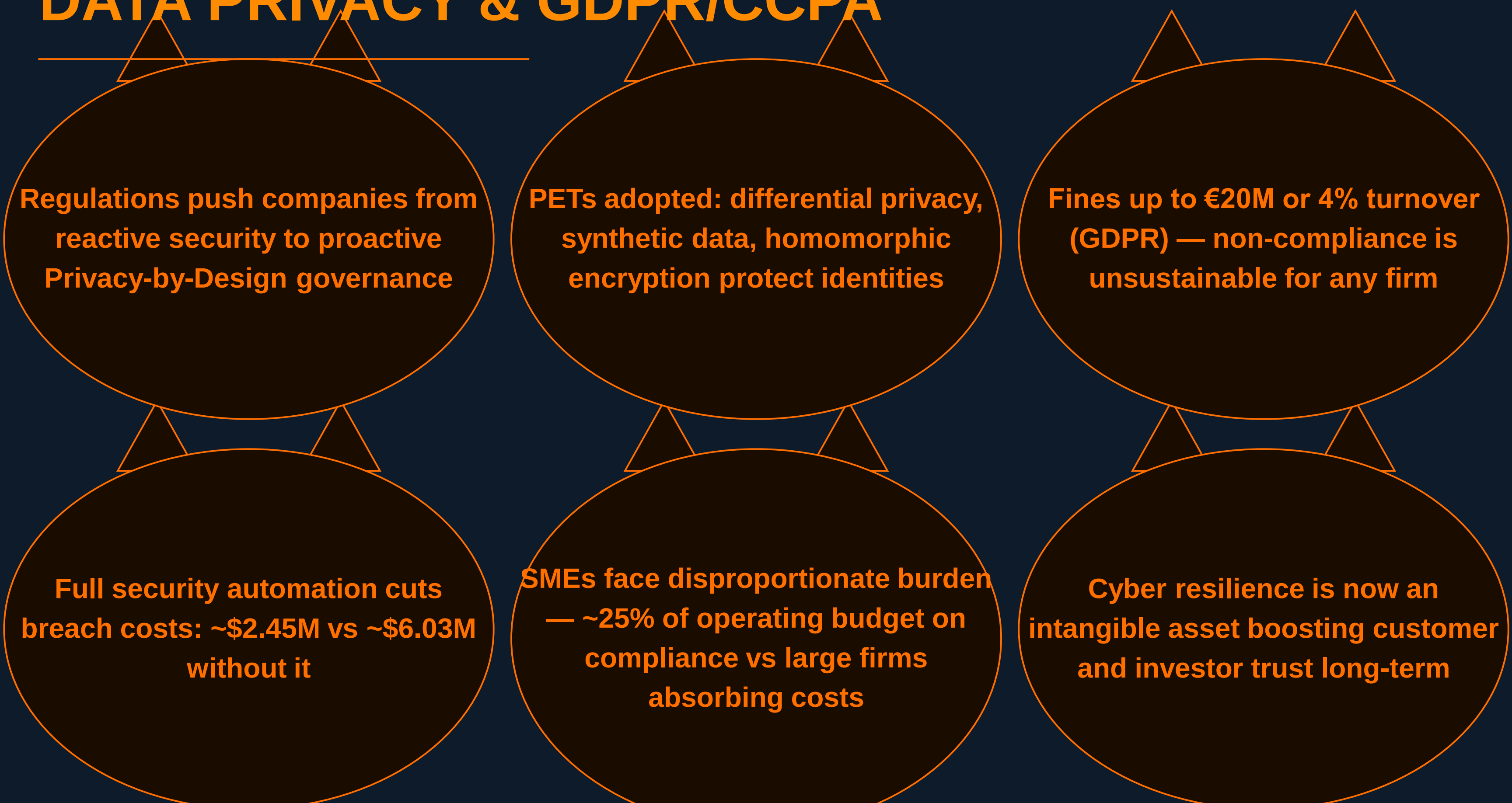
AI spear-phishing hits 54% click-through vs 12% for generic — 350% more effective than traditional

ML defenders detect subtle threats traditional rules miss — behavioral analysis flags anomalies

Automated response isolates compromised accounts in real time — reinforcement learning improves over time

Organizations must adopt Zero Trust, MFA, continuous training, and AI-driven tools to keep pace

DATA PRIVACY & GDPR/CCPA



Regulations push companies from reactive security to proactive Privacy-by-Design governance

PETs adopted: differential privacy, synthetic data, homomorphic encryption protect identities

Fines up to €20M or 4% turnover (GDPR) — non-compliance is unsustainable for any firm

Full security automation cuts breach costs: ~\$2.45M vs ~\$6.03M without it

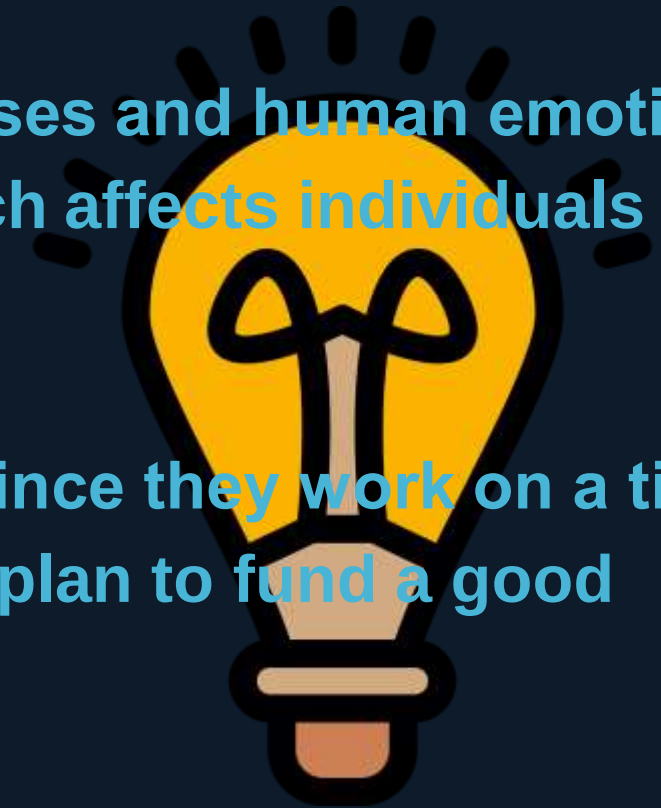
SMEs face disproportionate burden — ~25% of operating budget on compliance vs large firms absorbing costs

Cyber resilience is now an intangible asset boosting customer and investor trust long-term

CONCLUSIONS

Through intense research we realized the importance of cyber security in businesses.

- In the financial department, a strong security system saves companies billions of dollars in losses and helps them keep a trust-worthy image.
- In the humanitarian aspect, we realized phishing uses psychological weaknesses and human emotion to prey on users. Thus, strong security ideology needs to be incorporated, which affects individuals - positively - on a personal and professional level.
- We found out that most phishing attacks affect small enterprise businesses, since they work on a tight budget and with limited knowledge. Therefore, there needs to be a government plan to fund a good security system.
- We realized the impact AI has in both the phishing department and on the way it can be used to determine attacks and reduce them. Which is why companies need to invest in continuous training and AI tools that detect threats.
- Finally, we uncovered that privacy enhancing technologies enforce encryption that protects data identities and a good cyber resilience program boosts client trust.



Find our entire presentation on github!



THANK YOU

Questions?